DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 5

Lorentz force law, Biot-savart law, and ampère's law Date: 10 Mar 2025

1. A current I flows in a square loop of dimension $L \times L$. Find the magnetic field a distance z from the center of the loop in a direction perpendicular to the plane of the loop.
2. Find the force on a square loop placed as shown in the figure below (See Fig. 1). Assume that the straight wire is infinite in extent. Further assume that both the square loop and the straight wire carries a steady current I,

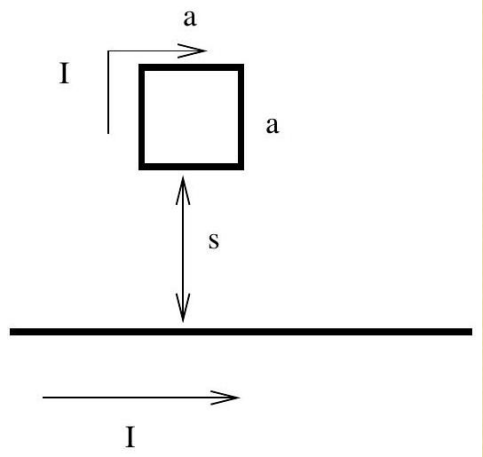


Figure 1: Figure for Problem 2

3. Similar to the above problem, find the force on a triangular loop (see Fig. 2). Similar conditions as elucidated for the above problem apply here too. Namely, the straight wire is infinite in extent and that both the triangular loop as well as the straight wire carries a steady current I.
4. A long cylindrical conductor of radius R has a cylindrical hole of radius b ($b < R$). The axis of the hole is parallel to the axis of the conductor. The remaining portion of the conductor has a uniform volume current density $\mathbf{J}$ parallel to the axis. Show that the magnetic field in the hole is uniform.
5. A thin isolated wire is bent in the form of a planar spiral consisting of a large number N of closely packed circular turns through which a steady current I flows. The inner radius of the spiral is a and the outer radius is b. Find the magnetic field $\mathbf{B}$ at the center of the spiral.
6. Two parallel conducting plates are at a distance d apart and have a potential difference ϕ_0 between them as shown in the figure 3. A uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ is also present in the region between the plates. Particles of charge q and mass m are released from rest at the point O. Find the equation of the trajectory of the particles and sketch this trajectory in the xy-plane.

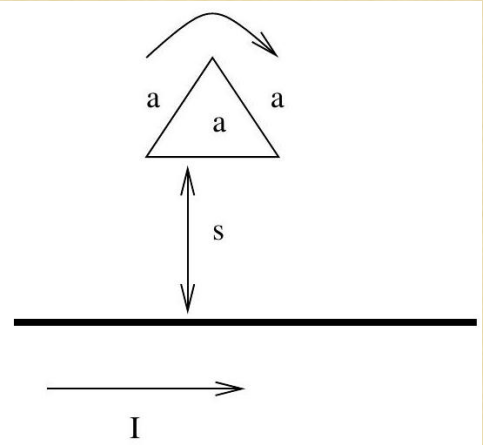


Figure 2: Figure for Problem 3

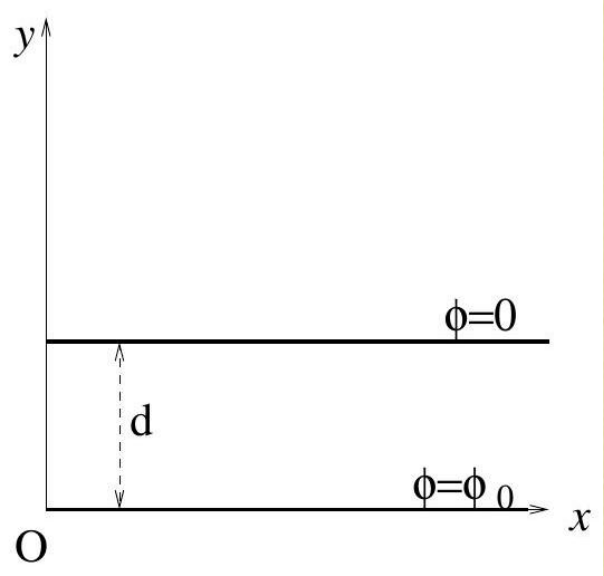


Figure 3: Figure for problem 6

7. **Understanding Helmholtz coils**: Two identical circular coils each having N turns and radius a are placed parallel to each other with a common axis. Their centres are separated by a distance $2d$, as shown in the figure 4. The width of each coil is negligible compared to a and d. Let O be the point midway between the centres of the two coils.

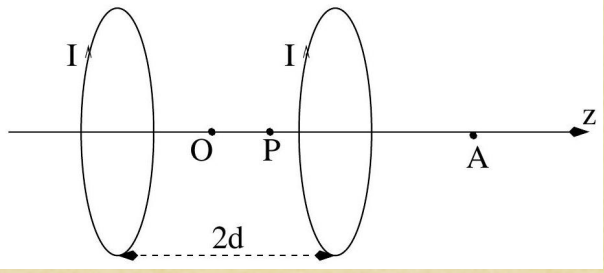


Figure 4: Figure for problem 7

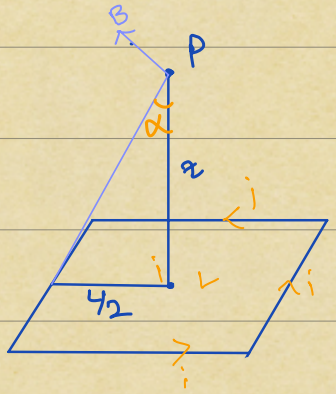
(a) As seen by an observer at A, the same current I flows through both coils in the clockwise sense. Find the magnetic induction $\mathbf{B}$ at a point P at a distance z from O.

(b) If the field is found to be uniform *correct to second order* in z, show that we must have $d = a/2$.

8. (a) Consider the infinite yz-plane. If a steady current of uniform surface current density $K_0\hat{\mathbf{z}}$ flows in this plane, find $\mathbf{B}$ at a point with coordinates (x, y, z). Sketch the magnetic field lines.

(b) Now suppose the infinite plane given by $x = L$ **also** carries the same uniform surface current density as the plane $x = 0$. Find $\mathbf{B}$ in the different regions corresponding to (i) $x < 0$, (ii) $0 < x < L$, (iii) $x > L$. Sketch the field lines.

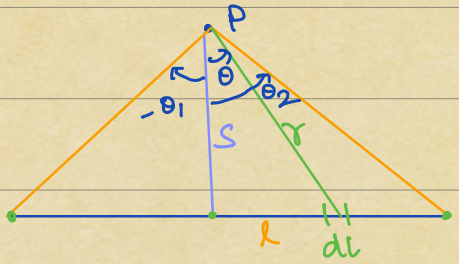
1. A current I flows in a square loop of dimension $L \times L$. Find the magnetic field a distance z from the center of the loop in a direction perpendicular to the plane of the loop.



$$B = 4B\sin\alpha$$

$$= 4 \times \frac{\mu_0 i}{2\sqrt{2}\pi s} \frac{l}{\sqrt{l^2+2z^2}} \times \frac{L/2 \times 2}{\sqrt{l^2+4z^2}}$$

$$\boxed{B = \frac{4\mu_0 i l^2}{\sqrt{2}\pi (l^2+4z^2)\sqrt{l^2+2z^2}}}$$



$$r = s\sec\theta$$

$$l = s.\tan\theta$$

$$dl = s\sec^2\theta\, d\theta$$

$$dB = \frac{\mu_0 i}{4\pi} \int \frac{s\sec^2\theta\, d\theta \times \sin(90+\theta)}{s^2\sec^2\theta}$$

$$\sin\theta_2 = \frac{l/2}{\sqrt{l^2/2+z^2}}$$

$$= \frac{\mu_0 i}{4\pi s}(\sin\theta_2 - \sin\theta_1) = \frac{\mu_0 i}{2\sqrt{2}\pi s} \frac{l}{\sqrt{l^2+2z^2}}$$

2. Find the force on a square loop placed as shown in the figure below (See Fig. 1). Assume that the straight wire is infinite in extent. Further assume that both the square loop and the straight wire carries a steady current I,

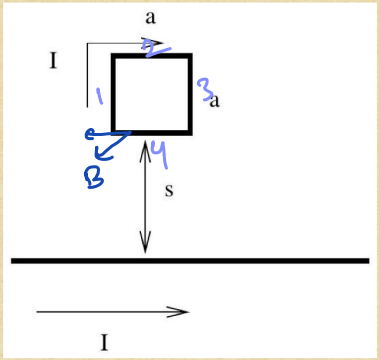


Figure 1: Figure for Problem 2

Surface 1,3 → cancels

Surface: 4

$$F = q(\vec{v} \times \vec{B}) = I(a)\left(\frac{\mu_0 I}{2\pi s}\right)\hat{j}$$

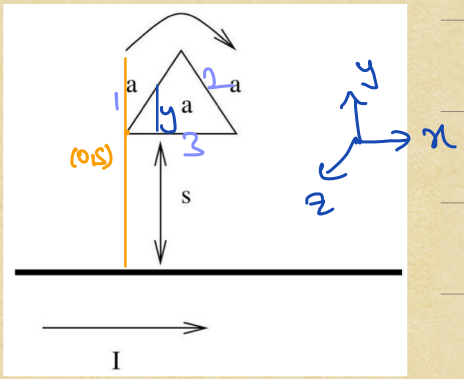
Surface: 2

$$I(a)\frac{\mu_0 I}{2\pi(s+a)}(-\hat{j})$$

$$\boxed{F = \frac{\mu_0 a I^2}{2\pi}\left(\frac{1}{s} - \frac{1}{s+a}\right)\hat{j}}$$

3. Similar to the above problem, find the force on a triangular loop (see Fig. 2). Similar conditions as elucidated for the above problem apply here too. Namely, the straight wire is infinite in extent and that both the triangular loop as well as the straight wire carries a steady current I.

$$F_3 = \frac{\mu_0 a I^2}{2\pi(s)}\,\hat{j}$$



for 1 and 2:

x-comp length cancels

$$\therefore \quad d\bar{l} = dx\,\hat{i} + dy\,\hat{j} \qquad B = \frac{\mu_0 i}{2\pi y}\,\hat{k}$$

$$df = (I\,d\bar{l} \times \bar{B}) = \frac{\mu_0 i^2}{2\pi y}(-dx\,\hat{j} + dy\,\hat{i})$$ ignore

$$\Rightarrow f_{left} = -\frac{\mu_0 i^2}{2\pi}\int_0^{a/2} \frac{dx}{y}\,\hat{j} \qquad (y = \sqrt{3}x + s)$$

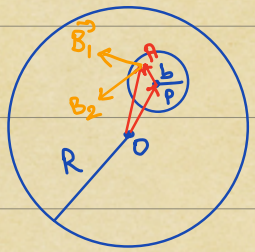
$$\Rightarrow f_{left} = -\frac{\mu_0 i^2}{2\pi\sqrt{3}} \ln\left(\frac{\frac{\sqrt{3}a}{2} + s}{s}\right)\hat{j}$$

$$f_{right} = -\frac{\mu_0 i^2}{2\pi}\int_{a/2}^{a} \frac{dx}{y}\,\hat{j} \qquad (y = -\sqrt{3}x + \sqrt{3}a + s)$$

$$f_{right} = \frac{\mu_0 i^2}{2\pi\sqrt{3}} \ln\left(\frac{s}{\frac{\sqrt{3}a}{2} + s}\right)\hat{j}$$

$$\boxed{\vec{F} = \left[\frac{\mu_0 i^2}{\pi\sqrt{3}} \ln\left(\frac{s}{\sqrt{3}a/2 + s}\right) + \frac{\mu_0 i^2 a}{2\pi s}\right]\hat{j}}$$

4. A long cylindrical conductor of radius R has a cylindrical hole of radius b ($b < R$). The axis of the hole is parallel to the axis of the conductor. The remaining portion of the conductor has a uniform volume current density $\mathbf{J}$ parallel to the axis. Show that the magnetic field in the hole is uniform.



B (cylinder):

$B \times 2\pi (OA) = \mu_0 (J \times \pi (OA)^2)$

$\Rightarrow B_1 = \frac{\mu_0 \vec{J} \times (\vec{OA})}{2}$

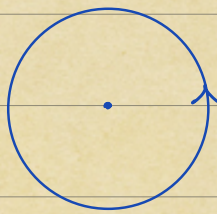
B (cavity)

$B \times 2\pi (PA) = \mu_0 (J \times \pi (PA)^2)$

$\Rightarrow B_2 = \frac{\mu_0 \vec{J} \times (\vec{PA})}{2}$

$\vec{B} = \vec{B_1} - \vec{B_2} = \frac{\mu_0 \vec{J} \times (\vec{OA} - \vec{PA})}{2} = \frac{\mu_0 \vec{J} \times (\vec{OP})}{2}$ = constant magnitude

5. A thin isolated wire is bent in the form of a planar spiral consisting of a large number N of closely packed circular turns through which a steady current I flows. The inner radius of the spiral is a and the outer radius is b. Find the magnetic field $\mathbf{B}$ at the center of the spiral.



$B = \frac{\mu_0}{4\pi_2} \frac{I (2\pi r)}{r^2} = \frac{\mu_0 I}{2r}$

number of small coils $= \frac{N}{b-a}$ $\quad K = \frac{NI}{b-a}$

$dB = \frac{\mu_0 dI}{2r} = \mu_0 \frac{N}{b-a} \frac{I \cdot dr}{2r} = \frac{\mu_0 N i}{2(b-a)} \ln\left(\frac{b}{a}\right)$

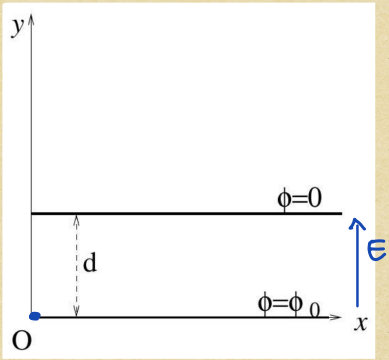
outside plane

6. Two parallel conducting plates are at a distance d apart and have a potential difference ϕ_0 between them as shown in the figure 3. A uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ is also present in the region between the plates. Particles of charge q and mass m are released from rest at the point O. Find the equation of the trajectory of the particles and sketch this trajectory in the xy-plane.

$\nabla^2\phi = 0$

$\frac{d^2\phi}{dy^2} = 0$

$\Rightarrow \phi = ay + b$



$E = \frac{\phi_0}{d}$

$b = \phi_0 \quad a = -\frac{\phi_0}{d}$

$\Rightarrow \phi = -\frac{\phi_0}{d}y + \phi_0 \qquad E = -\nabla\phi = \frac{\phi_0}{d}\hat{y}$

$v = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

$\vec{F}_B = q(v_x\hat{i} + v_y\hat{j}) \times (B\hat{k})$

$\vec{F}_B = qv_xB(-\hat{j}) + qv_yB\hat{i}$

$\vec{F}_e = \frac{q\phi_0}{d}\hat{j} \Rightarrow F = \left(-qv_xB + \frac{q\phi_0}{d}\right)\hat{j} + qv_yB\hat{i}$

$a_x = \left(\frac{qB}{m}\right)v_y \qquad a_y = \frac{q}{m}\left(\frac{\phi_0}{d} - Bv_x\right)$

$\Rightarrow \frac{dv_x}{dt} = \left(\frac{qB}{m}\right)v_y \qquad \frac{dv_y}{dt} = \frac{q}{m}\left(\frac{\phi_0}{d} - Bv_x\right)$

$\frac{d^2v_y}{dt} = -\frac{qB}{m}\left(\frac{qB}{m}v_y\right) \Rightarrow v_y = A\sin\left(\frac{qB}{m}t\right)$

$\frac{dv_x}{dt} = \frac{qB}{m} \cdot A\sin\left(\frac{qB}{m}t\right)$

$$\Rightarrow v_x = A\left[-\cos\frac{qBt}{m} + 1\right]$$

$$\Rightarrow x = \frac{mA}{qB}\left[-\sin\frac{qBt}{m} + \frac{qBt}{m}\right]$$

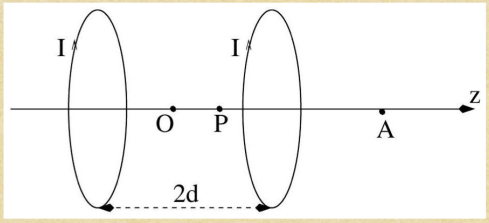
$$y = \frac{mA}{qB}\left[-\cos\frac{qBt}{m} + 1\right]$$

For Amplitude:

At $t=0$, $a_y = \frac{qQ_0}{md}$

$$\frac{dv_y}{dt} = \frac{AqB}{m} = \frac{qQ_0}{md} \Rightarrow A = \frac{Q_0}{Bd}$$

7. **Understanding Helmholtz coils**: Two identical circular coils each having N turns and radius a are placed parallel to each other with a common axis. Their centres are separated by a distance $2d$, as shown in the figure 4. The width of each coil is negligible compared to a and d. Let O be the point midway between the centres of the two coils.



(a) As seen by an observer at A, the same current I flows through both coils in the clockwise sense. Find the magnetic induction **B** at a point P at a distance z from O.

(b) If the field is found to be uniform *correct to second order* in z, show that we must have $d = a/2$.

$Z \propto d$: $\frac{\mu_0}{4\pi}\frac{I \times 2\pi a^2 N}{(a^2+(z+d)^2)^{3/2}}(\hat{z}) +$

$\frac{\mu_0}{4\pi}\frac{I \times 2\pi a^2 N}{(a^2+(z-d)^2)^{3/2}}(\hat{z})$

$$B = \frac{\mu_0 I a^2 N}{2}\left[\frac{1}{(a^2+(z-d)^2)^{3/2}} + \frac{1}{(a^2+(z+d)^2)^{3/2}}\right](\hat{z})$$

$$B(z) = B(0) + B'(0)z + B''(0)\frac{z^2}{2!} + B'''(0)\frac{z^3}{3!} + \cdots$$

$$\frac{dB}{dz} = \frac{-3}{2}\frac{2(z-d)}{[a^2+(z-d)^2]^{5/2}} - \frac{3}{2}\frac{2(z+d)}{[a^2+(z+d)^2]^{5/2}} = 0 \quad @\ z=0$$

$$\frac{d^2B}{dz^2} = -3\frac{[a^2+(z-d)^2]^{5/2} - (z-d)\,5/2\,[a^2+(z-d)^2]^{3/2}\,2(z-d)}{(a^2+(z-d)^2)^5}$$

$$-3\frac{[a^2+(z+d)^2]^{5/2} - (z+d)\,5/2\,[a^2+(z+d)^2]^{3/2}\,2(z+d)}{(a^2+(z+d)^2)^5}$$

$$=$$

$$-[a^2+d^2]^{5/2} + d\,(5/2\,[a^2+d^2]^{3/2})(-2d)$$

$$-[a^2+d^2]^{5/2} + d\,(5/2\,[a^2+d^2]^{3/2})(2d)$$

$$= \frac{-2[a^2+d^2]^{5/2} + 10d^2(a^2+d^2)^{3/2}}{(a^2+d^2)^5}$$

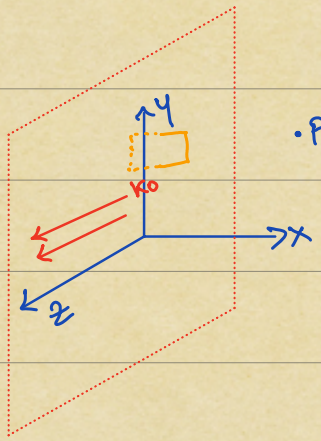
$$= \frac{-2[a^2+d^2] + 10d^2}{(a^2+d^2)^{7/2}} \qquad -2a^2+8d^2=0$$

$$\Rightarrow d = a/2$$

8. (a) Consider the infinite yz-plane. If a steady current of uniform surface current density $K_0\hat{\mathbf{z}}$ flows in this plane, find $\mathbf{B}$ at a point with coordinates (x, y, z). Sketch the magnetic field lines.

   (b) Now suppose the infinite plane given by $x = L$ **also** carries the same uniform surface current density as the plane $x = 0$. Find $\mathbf{B}$ in the different regions corresponding to (i) $x < 0$, (ii) $0 < x < L$, (iii) $x > L$. Sketch the field lines.

(a)



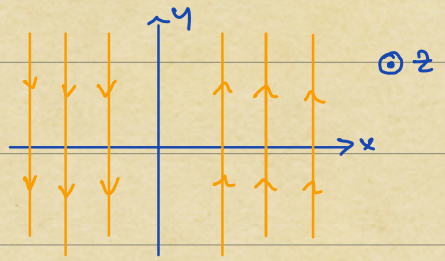
$\cdot P(x,y,z)$

$Bx2l = \mu_0(Kxl)$

$\Rightarrow B = \frac{\mu_0 K}{2}$

$x>0 \Rightarrow B: \hat{y}$

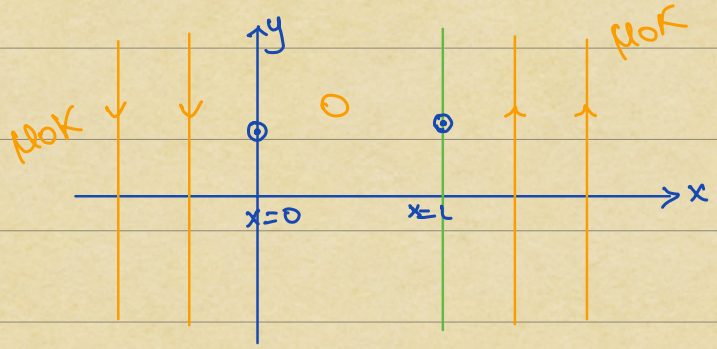
$x<0 \Rightarrow B: -\hat{y}$



(b) region outside: $\mu_0 K$

$x<0: -\hat{y}$ $x>L: \hat{y}$

region inside: 0



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PH1020 Physics II
Problem Set 5 - Solutions

Lorentz force law, Biot-savart law, and ampère's law Date: 10 Mar 2025

1. Problem-1

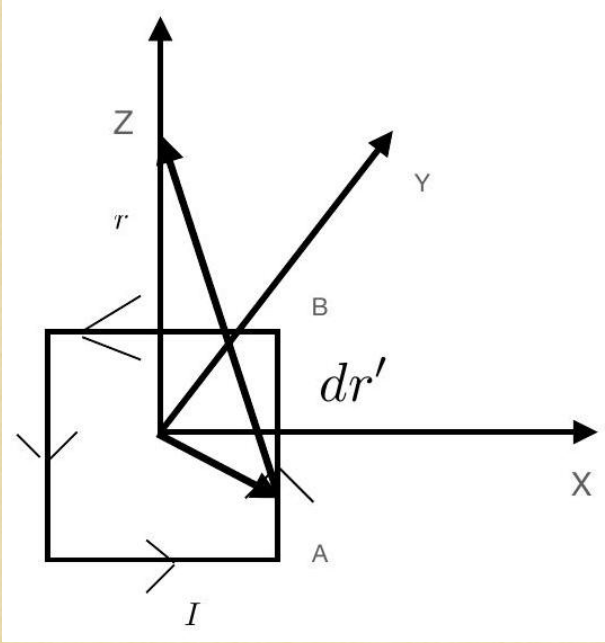
Question: A current I flows in a square loop of dimension $L \times L$. Find the magnetic field a distance z from the center of the loop in a direction perpendicular to the plane of the loop.

Solution:

Use Biot-Savart law:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int d\mathbf{r}' \times \frac{(\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}$$

where $d\mathbf{r}'$ is an element along the current. For the segment designated AB, (see figure below) :



$$\mathbf{r}' = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} = x\hat{\mathbf{x}} + \frac{L}{2}\hat{\mathbf{y}}$$

, and the element

$$d\mathbf{r}' = dx\hat{\mathbf{x}}$$

Thus,

$$d\mathbf{r}' \times (\mathbf{r} - \mathbf{r}') = -\left(z\hat{\mathbf{z}} - x\hat{\mathbf{x}} - \frac{L}{2}\hat{\mathbf{y}}\right) \times dx\hat{\mathbf{x}} = -zdx\hat{\mathbf{y}} - \frac{L}{2}\hat{\mathbf{z}}$$

From symmetry, one can argue that there are no components in the $x-y$ plane of the current. Thus, only the $\hat{\mathbf{z}}$ component survives. Thus, the magnetic field in the $\hat{\mathbf{z}}$ due to the line segment designated AB is given by:

$$B_{\rm AB} = \frac{\mu_0 I}{4\pi}\int_{\mathbf{L/2}}^{\mathbf{-L/2}} \frac{-\frac{\mathbf{L}}{2}dx}{\left(\mathbf{L}^2/4 + z^2 + x^2\right)^{3/2}}$$

The net field due to all the segments is thus $4\times$ the above result. Hence, we get the net field

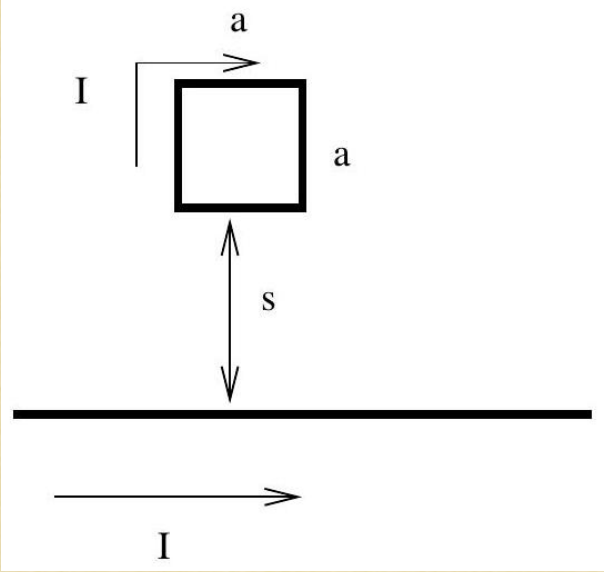
$$B_z = \frac{\mu_0 I L}{\pi}\int_{0}^{\mathbf{L/2}} \frac{dx}{\left(\mathbf{L}^2/4 + z^2 + x^2\right)^{3/2}}$$

Performing the integral we get

$$\mathbf{B} = \frac{4\mu_0 I L^2}{\pi\left(4z^2+L^2\right)\sqrt{4z^2+2L^2}}\hat{\mathbf{z}}.$$

2. Problem-2

Question: Find the force on a square loop placed as shown in the figure below (See Fig.below). Assume that the straight wire is infinite in extent. Further assume that both the square loop and the straight wire carries a steady current I,



Solution:

The force on the two sides cancel each other. Thus, one needs to look at the force contributions from the bottom and top segments of the square loop: At the bottom, the field is

$$B = \frac{\mu_0 I}{2\pi s}\hat{\mathbf{z}}.$$

. To find the force $F = I\int d\boldsymbol{\ell}\times\mathbf{B}$ gives

$$F_{\rm lower} = \frac{\mu_0 I^2}{2\pi s}\int_0^a \left(dx\left(-\hat{\mathbf{x}}\right)\times\hat{\mathbf{z}}\right) = \frac{\mu_0 I^2 a}{2\pi s}\hat{\mathbf{y}}$$

The force on the upper line segment, F_{upper} of the loop can also be found in similar fashion. However in this case the magnetic field felt by the upper segment of the square is

$$B_{\text{upper}} = \frac{\mu_0 I}{2\pi(s+a)}\hat{\mathbf{z}}$$

Once again,

$$F_{\text{upper}} = \frac{\mu_0 I^2}{2\pi(s+a)}\int_0^a (dx\,(\hat{\mathbf{x}}) \times \hat{\mathbf{z}}) = -\frac{\mu_0 I^2 a}{2\pi(s+a)}\hat{\mathbf{y}}$$

. Thus, the net force felt on the square loop is thus

$$F = \frac{\mu_0 I^2 a}{2\pi}\left(\frac{1}{s} - \frac{1}{s+a}\right)\hat{\mathbf{y}} = \frac{\mu_0 I^2 a^2}{2\pi s(s+a)}\hat{\mathbf{y}}.$$

3. Problem-3

Question: Similar to the above problem, find the force on a triangular loop (see Fig.). Similar conditions as elucidated for the above problem apply here too. Namely, the straight wire is infinite in extent and that both the triangular loop as well as the straight wire carries a steady current I.

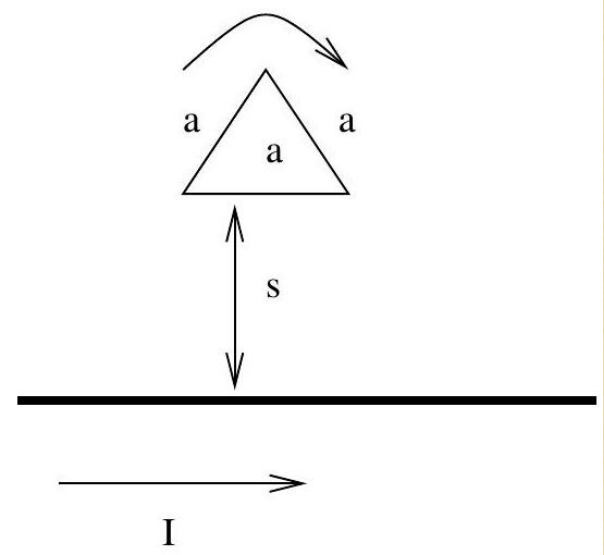


Figure 1: Figure for Problem 3

Solution:
For the lower line segment of the triangle we can borrow the result from the previous problem. Thus we have

$$F_{\text{lower}} = \frac{\mu_0 I^2 a}{2\pi s}\hat{\mathbf{y}}.$$

Now, we need to find the force on the other two segments. To do so, we let us look at the incremental force $dF = d\ell \times B$ felt by a segment $d\ell$:

$$dF = I(d\boldsymbol{\ell} \times B) = I\,(dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}) \times \frac{\mu_0 I}{2\pi y}\hat{\mathbf{y}} = \frac{\mu_0 I^2}{2\pi y}\,(-dx\hat{\mathbf{y}} + dy\hat{\mathbf{x}})$$

The wire on the left hand side has an x-component of the force that cancels exactly with the x-component coming from the leg of the triangle placed on the left hand side. Thus, the force on the left leg is

$$F_{\text{left}} = \frac{\mu_0 I^2}{2\pi} \int_0^{a/2} \frac{-dx}{y} \hat{\mathbf{y}}.$$

Please note y is dependent on x. From the figure this can easily be worked out to be $y = \sqrt{3}x + s$. Thus, we get

$$F_{\text{left}} = \frac{\mu_0 I^2}{2\pi} \int_0^{a/2} \frac{-dx}{\sqrt{3}x + s} \hat{\mathbf{y}} = \frac{\mu_0 I^2}{2\pi} \left(\frac{-\hat{\mathbf{y}}}{\sqrt{3}} \ln \frac{s + a\sqrt{3}}{s} \right)$$

Now, for the right leg we follow the same strategy. Thus, we get

$$F_{\text{left}} = \frac{\mu_0 I^2}{2\pi} \int_{a/2}^{a} \frac{-dx}{y} \hat{\mathbf{y}}.$$

However in this case the constraint equation connecting y and x is given by $y = -\sqrt{3}x + s + a\sqrt{3}$

$$F_{\text{right}} = \frac{\mu_0 I^2}{2\pi} \int_{a/2}^{a} \frac{-dx}{-\sqrt{3}x + s + a\sqrt{3}} \hat{\mathbf{y}} = \frac{\mu_0 I^2}{2\pi} \left(\frac{\hat{\mathbf{y}}}{\sqrt{3}} \ln \frac{s}{s + a\sqrt{3}/2} \right)$$

. Thus, the net force $F = F_{\text{lower}} + F_{\text{left}} + F_{\text{right}}$ which gives

$$F = \frac{\mu_0 I^2}{2\pi} \left(\frac{a}{s} + \frac{2}{\sqrt{3}} \ln \left(\frac{s}{s + a\sqrt{3}/2} \right) \right) \hat{\mathbf{y}}.$$

4. Problem-4

Question: A long cylindrical conductor of radius R has a cylindrical hole of radius b $(b < R)$. The axis of the hole is parallel to the axis of the conductor. The remaining portion of the conductor has a uniform volume current density $\mathbf{J}$ parallel to the axis. Show that the magnetic field in the hole is uniform.

Solution:

First indicate the magnetic field $\mathbf{B}$ due to a long straight cylindrical conductor with uniform current density $\mathbf{J}$, parallel to the axis. For a circular path of radius $\rho(\rho < R)$ in a plane normal to the axis of the conductor, the current enclosed is $\pi\rho^2 J$.$\mathbf{B}$ is tangential to the circle everywhere and of the same magnitude because of the cylindrical symmetry. (Recall the argument for the case of magnetic field due to an infinitely long wire, done in class) Therefore, for $\rho < R$

$$\begin{aligned} \oint \mathbf{B} \cdot \mathbf{dl} &= B(\rho)(2\pi\rho) \\ &= \mu_0 \pi \rho^2 J \\ \implies B(\rho) &= \frac{\mu_0 J \rho}{2} \end{aligned}$$

In vector form,

$$\mathbf{B}(\rho < R) = \frac{\mu_0}{2} (\mathbf{J} \times \boldsymbol{\rho})$$

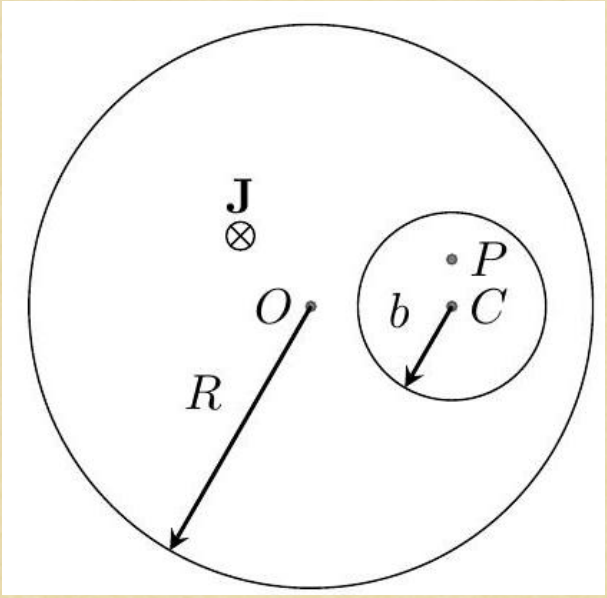


Figure 2: Hole in a cylindrical conductor

For $\rho > R$

$$\oint \mathbf{B} \cdot \mathbf{dl} = \mu_0 \pi R^2 J$$

$$\implies \mathbf{B}(\rho \geq R) = \frac{\mu_0 R^2}{2\rho^2}(\mathbf{J} \times \boldsymbol{\rho}).$$

Now, we consider the case of a cavity in this cylinder. This cavity has zero current density. C is the center of the cavity. $\mathbf{B}($ at $P)$ if the entire cylinder had a current density is

$$\frac{\mu_0}{2}(\mathbf{J} \times \mathbf{OP})$$

$\mathbf{B}($ at $P)$ if the cavity alone had a current density $\mathbf{J}$ is

$$\frac{\mu_0}{2}(\mathbf{J} \times \mathbf{CP})$$

Therefore,

$$\text{net}\, \mathbf{B}(\text{ at P}) = \frac{\mu_0}{2}(\mathbf{J} \times (\mathbf{OP} - \mathbf{CP}))$$

$$\mathbf{B} = \frac{\mu_0}{2}(\mathbf{J} \times \mathbf{OC}).$$

The dependence on P has totally disappeared and $\mathbf{B}$ is uniform within the hole at all points and its direction is normal to OC.

5. Problem-5

Question: A thin isolated wire is bent in the form of a planar spiral consisting of a large number N of closely packed circular turns through which a steady current I flows. The inner radius of the spiral is a and the outer radius is b. Find the magnetic field $\mathbf{B}$ at the center of the spiral.

Solution:

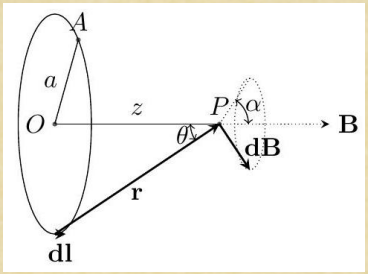


Figure 3: The magnetic field due to a circular loop

This problem is a simple application of Biot-Savart's law. The equation of the spiral coil is

$$\varrho(\varphi) = (b-a)\frac{\varphi}{2\pi N} + a, \quad \varphi \in [0, 2\pi N].$$

An infinitesimal element has $\mathbf{dl} = d\varrho(\varphi)\hat{\varrho} + \varrho(\varphi)d\varphi\hat{\varphi}$. At the centre of the coil, one has

$$\mathbf{dB} = \frac{\mu_0}{4\pi} i\mathbf{dl} \times \frac{-\hat{\varrho}}{\varrho^2} = \frac{\mu_0}{4\pi}\frac{id\varphi}{\varrho}\hat{\mathbf{z}}$$

Integrating over the coil, we get

$$\mathbf{B} = \frac{\mu_0 i}{4\pi}\hat{\mathbf{z}}\int_0^{2\pi N}\frac{d\varphi}{\varrho(\varphi)}$$
$$\mathbf{B} = \frac{\mu_0 iN}{2(b-a)}\ln\frac{b}{a}\hat{\mathbf{z}}.$$

Alternate Solution

This problem can be solved using the expression for the magnetic field of a single loop (carrying a steady current), at the center of the loop. Let us determine the (more general case of the) magnetic field at a distance z along the axis of a circular current carrying loop.

The magnetic field, **dB**, the magnetic field due to the element **dl** at P will clearly be perpendicular to **dl** and **r**. The angle between **dl** and **r** is $\pi/2$. Biot-Savart's law gives

$$dB = \frac{\mu_0}{4\pi}\frac{idlr}{r^3} = \frac{\mu_0}{4\pi}\frac{idl}{(z^2+a^2)},$$

where a is the radius of the loop. To obtain **B**, we integrate **dl** over the loop and correspondingly **dB** sweeps out a cone as shown, with semi-vertical angle $\alpha = \left(\frac{\pi}{2} - \theta\right)$. Further, only the component of **B** along the z axis remains, as the other component cancels out. $\left(\cos\alpha = \sin\theta = \frac{a}{\sqrt{a^2+z^2}}\right)$

$$\therefore |\mathbf{B}| = \frac{\mu_0 i}{4\pi}\int\frac{\mathrm{dl}}{r^2}\cos\alpha = \frac{\mu_0 i}{4\pi r^2}\cos\alpha(2\pi a) = \frac{\mu_0 i a^2}{2\left(a^2+z^2\right)^{3/2}}$$
$$\mathbf{B} = \frac{\mu_0 i a^2}{2\left(a^2+z^2\right)^{3/2}}\hat{\mathbf{z}}.$$

The field at the center of the loop has magnitude $\frac{\mu_0 i}{2a}$. Now, in this problem, the contribution from one turn of the planar spiral to the field at the center is $\frac{\mu_0 i}{2\rho}$ (ρ is the

radius of the turn). Therefore, the field at the center from all N turns is

$$\int \frac{\mu_0 i}{2\varrho} dN = \int \frac{\mu_0 i}{2\varrho} \frac{N}{(b-a)} d\varrho,$$

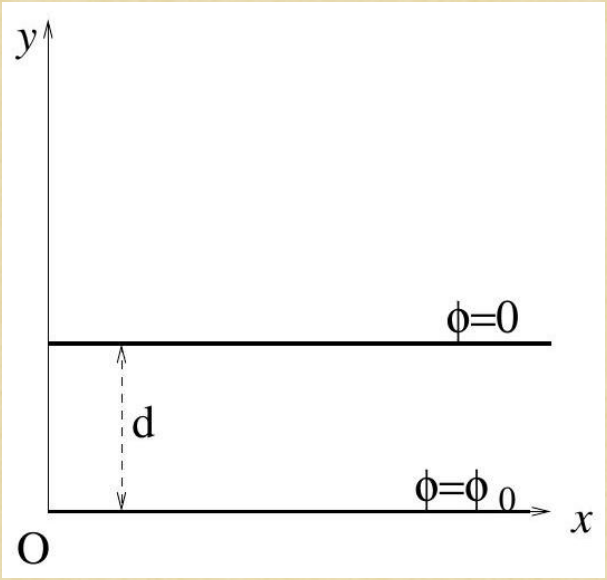
where dN is the number of the turns that lie in the interval $(\varrho, \varrho + d\varrho)$.

$$\begin{aligned} \therefore \mathbf{B} &= \frac{\mu_0 i}{2} \frac{N}{(b-a)} \int_a^b \frac{d\varrho}{\varrho} \\ &= \frac{\mu_0 i}{2} \frac{N}{(b-a)} \ln\left(\frac{b}{a}\right) \hat{\mathbf{z}} \end{aligned}$$

We can think of the above computation as arising due to a surface current density $\mathbf{K} = K(\varrho)\hat{\varphi}$ with $K(\varrho) = \frac{N}{(b-a)}$.

6. Problem-6

Question: Two parallel conducting plates are at a distance d apart and have a potential difference ϕ_0 between them as shown in the figure. A uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ is also present in the region between the plates. Particles of charge q and mass m are released from rest at the point O. Find the equation of the trajectory of the particles and sketch this trajectory in the xy-plane.



Solution:

Determining the trajectory of a charged particle

$$\mathbf{B} = B\hat{\mathbf{z}} \quad \text{and} \quad \mathbf{E} = E\hat{\mathbf{y}} = \frac{\phi_0}{d}\hat{\mathbf{y}}$$

Therefore, the force is along the y axis ($q\mathbf{E}$) due to the electric field and is in the xy plane ($\mathbf{v} \times \mathbf{B}$), due to $\mathbf{B}$. The particle has a velocity

$$\begin{aligned} \mathbf{v} &= \dot{x}\hat{\mathbf{x}} + \dot{y}\hat{\mathbf{y}} \\ \mathbf{F} &= m\ddot{x}\hat{\mathbf{x}} + m\ddot{y}\hat{\mathbf{y}} \end{aligned}$$

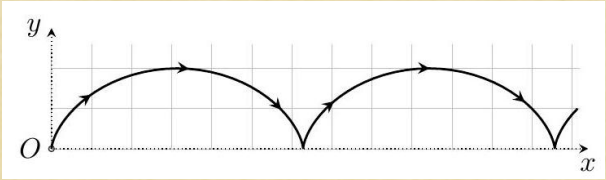


Figure 4: Cycloid

Equate $\mathbf{F}$ to $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ component wise.

$$\ddot{x} = \omega\dot{y}$$
$$\ddot{y} = -\omega\dot{x} + \frac{qE}{m}$$

with $\omega = \frac{qB}{m}$ being the cyclotron frequency. Differentiate equations (4) and (5) to get the uncoupled equations

$$\dddot{x} = -\omega^2\dot{x} + \frac{\omega qE}{m}$$
$$\dddot{y} = -\omega\ddot{x} = -\omega^2\dot{y}$$

Impose boundary conditions and solve. At $t = 0$,

$$y = 0, \quad \dot{y} = 0, \quad \ddot{y} = \frac{qE}{m}.$$

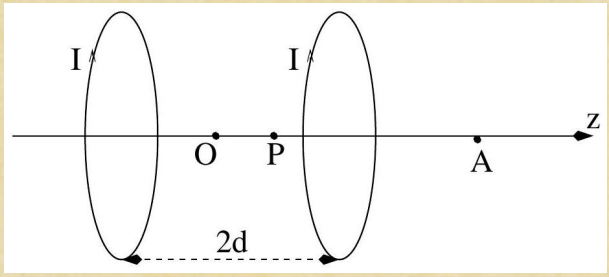
The solution for $y(t)$ is $y(t) = \frac{E}{\omega B}(1 - \cos(\omega t))$. Therefore,

$$x = \frac{E}{\omega B}(\omega t - \sin\omega t)$$
$$\text{and } y = \frac{E}{\omega B}(1 - \cos(\omega t)).$$

Equations (8) and (9) are the parametric equations for a cycloid, shown below in Figure 3

7. Problem-7

Question: Understanding Helmholtz coils: Two identical circular coils each having N turns and radius a are placed parallel to each other with a common axis. Their centres are separated by a distance $2d$, as shown in the figure. The width of each coil is negligible compared to a and d. Let O be the point midway between the centres of the two coils.



(a) As seen by an observer at A, the same current I flows through both coils in the clockwise sense. Find the magnetic induction $\mathbf{B}$ at a point P at a distance z from O.

(b) If the field is found to be uniform *correct to second order* in z, show that we must have $d = a/2$.

Solution:

(a) Using the expression for the magnetic field along the axis of a single coil given in Eq. (1) after suitably accounting for the direction of the current as well as the number of turns, we obtain

$$\mathbf{B}_1 \text{ at } P \text{ due to coil } 1 = -\frac{\mu_0 NI}{4\pi}\frac{2\pi a^2}{\left[a^2+(d+z)^2\right]^{3/2}}\hat{\mathbf{z}}$$

$$\mathbf{B}_2 \text{ at } P \text{ due to coil } 2 = -\frac{\mu_0 NI}{4\pi}\frac{2\pi a^2}{\left[a^2+(d-z)^2\right]^{3/2}}\hat{\mathbf{z}}$$

$$\therefore \mathbf{B}(z) = \mathbf{B}_1 + \mathbf{B}_2 = -\frac{\mu_0 NIa^2}{2}\hat{\mathbf{z}}\left(\frac{1}{\left[a^2+(d+z)^2\right]^{3/2}} + \frac{1}{\left[a^2+(d-z)^2\right]^{3/2}}\right)$$

(b) Expand $\mathbf{B}(z)$ in terms of $\mathbf{B}(0)$ as a Taylor series in z (for small z), we get

$$\mathbf{B}(z) = \mathbf{B}(0)\left(1 + \frac{3\left(-a^2+4d^2\right)}{2\left(a^2+d^2\right)^2}z^2 + \frac{15\left(a^4-12a^2d^2+8d^4\right)}{8\left(a^2+d^2\right)^4}z^4 + \mathcal{O}\left(z^6\right)\right)$$

The odd power of z do not appear as the magnetic field is an even function of z. When $4d^2 = a^2$ or $d = \frac{a}{2}$, the term proportional to z^2 vanishes and we obtain

$$\mathbf{B}(z)|_{d=\frac{a}{2}} = \mathbf{B}(0)\left(1 - \frac{144z^4}{125a^4} + \mathcal{O}\left(z^6\right)\right)$$

Therefore, when $d = \frac{a}{2}, \mathbf{B}$ is uniform to the third power of z. This setup is called Helmholtz coils and is used to provide an approximately uniform magnetic field $\mathbf{B}$ at O.

8. Problem-8

Question:

(a) Consider the infinite yz-plane. If a steady current of uniform surface current density $K_0\hat{\mathbf{z}}$ flows in this plane, find $\mathbf{B}$ at a point with coordinates (x, y, z). Sketch the magnetic field lines.

(b) Now suppose the infinite plane given by $x = L$ **also** carries the same uniform surface current density as the plane $x = 0$. Find $\mathbf{B}$ in the different regions corresponding to (i) $x < 0$, (ii) $0 < x < L$, (iii) $x > L$. Sketch the field lines.

Solution:

(a) The expression for $|\mathbf{B}|$ at a distance x away from the sheet has already been obtained in class. The direction of $|\mathbf{B}|$ is along $\hat{e}_y$ and has a magnitude $\frac{\mu_0 K}{2}$. $\mathbf{B}$ is therefore uniform.

(b) Using superposition, we see that In between the sheets,

$$\mathbf{B} = \frac{\mu_0 K}{2}\hat{\mathbf{y}} - \frac{\mu_0 K}{2}\hat{\mathbf{y}} = \mathbf{0}$$

whereas for the region outside the sheets, $\mathbf{B}$ has magnitude $\mu_0 K$.

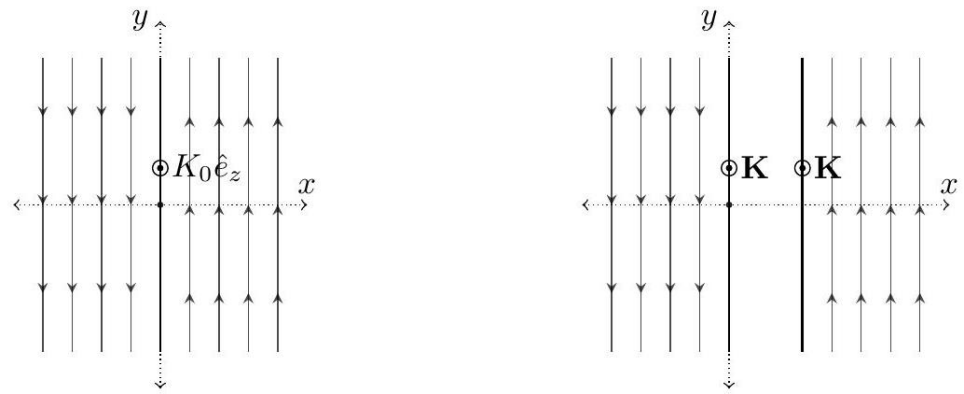


Figure 5: To the left is the magnetic field lines due to a single sheet and to the right is the magnetic field lines due to two parallel sheets.

[1]

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